Day - 4

Aptitude (Time and work)

1) A's 1 day work = 1/10
B's = 1/15

Together ⇒ $\frac{1\times3}{10\times3} + \frac{1\times2}{15\times2} = \frac{3\times2}{30}$

$= 5/30 = 1/6$

2) A = 1/6, B = 1/8, C = 1/12

$\frac{1\times4}{6\times4} + \frac{1\times3}{8\times3} + \frac{1\times2}{12\times2}$

$= \frac{4+3+2}{24} = \frac{9}{24} = \frac{3}{8}$ Time (Required);

$= \frac{8}{3} = 2\frac{2}{3}$ days

3) $A = \frac{1}{12}$

A in 4 days = $4 \times \frac{1}{12} = \frac{1}{3}$

Balance =

$1 - \frac{1}{3} = \frac{2}{3}$

A+B 4 days

$(A+B) = \frac{2/3}{4} = \frac{1}{6}$

$A + B = \frac{1}{6}$, $A = \frac{1}{12}$

$B = \frac{1}{6} - \frac{1}{12} = \frac{1}{12}$

4)

$5 \times 8 = 40$ ~~total~~

cost = x

$x \times 4 = 40$

$x = 10$

5) Filling pipe $= 1/6$

Emptying pipe $= \frac{1}{8}$

$\frac{1}{6} - \frac{1}{8} = \frac{4-3}{24} = \frac{1}{24}$

Time = 24 hours.

Block - 2 (Verbal ability)

Reading passage

1) Optimized systems are important because they,

Improve performance

Reduce resource usage

Handle large data efficiently

2) Scalability is a system's ability to;

→ Handle increasing users, data and workload.

→ without slowing down or crashing

3) Amazon

Google

Microsoft.

Facebook.

Grammar correction:

1) He didn't knew the answer.

He didn't know the answer

2) she is more smarter than him

she is smarter than him.

3) They has completed the task.

They have completed the task

Block 3:

An array is a data structure and stores multiple values of the same data type in a single variable

Each value in the array is stored at a specific position called an index, which helps us access elements easily

In real life, you can think of an array like a row of lockers, where each locker has a number and stores one item

Use case:-

Arrays are used in applications like storing user data, prices in shopping apps or handling large datasets efficiently